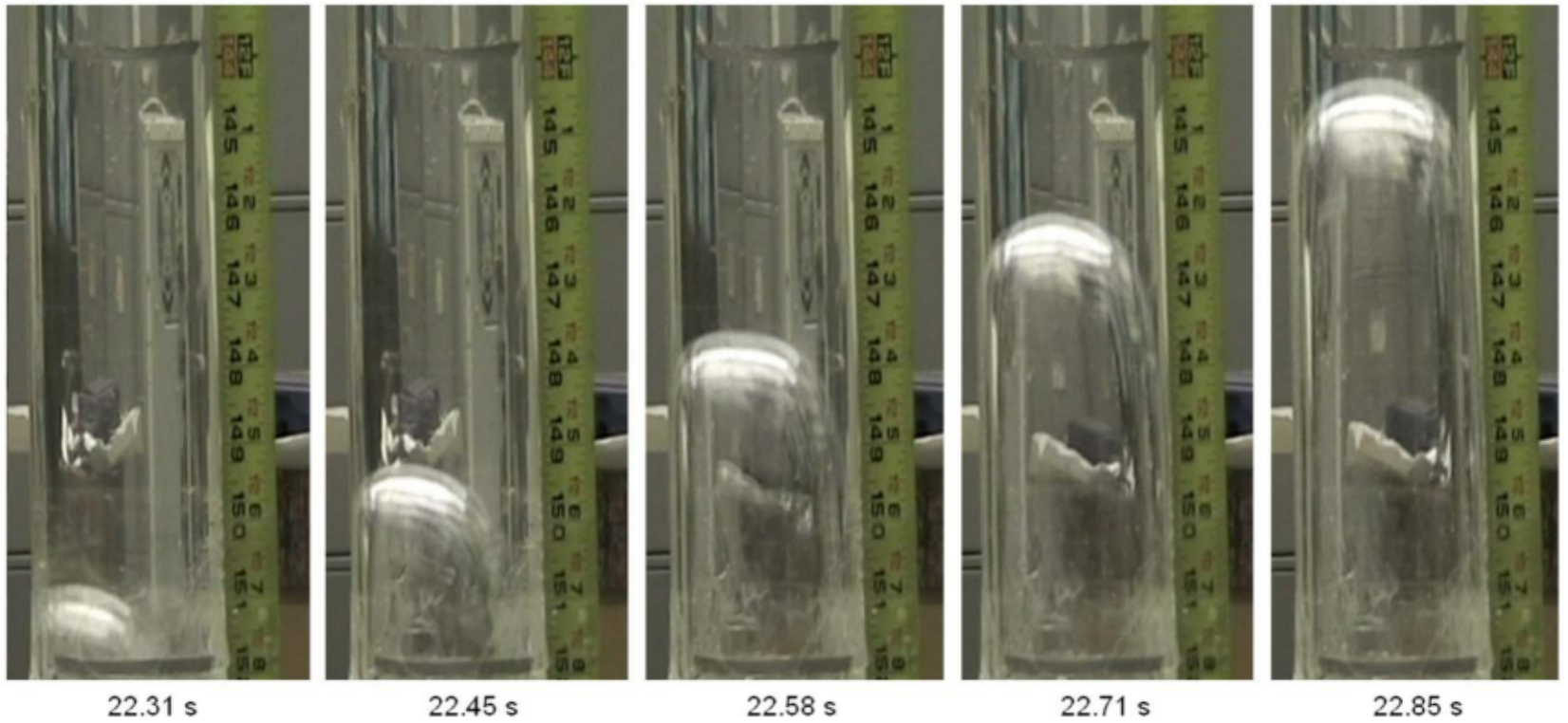


(a) Experiment: Vasconcelos and Wright (2011), Fig. 4



(b) Present two-fluid model at the same 0.14 s spacing

Case A -- model riser snapshots, 0.14 s apart (cf. paper Fig.4)
 gas-core width = local two-fluid gas fraction α_g (no thresholding); grey above blue line = atmosphere over the free surface;
 red dashes = pocket gas front Y_{int} , blue line = free surface Y_{fs}

